

Eight-Channel Data Acquisition & Logging System

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In environments like factories, power plants and transformers in electricity substations, controlling temperature to a safe value is important. Supervisory and control systems are used to monitor the temperature

port of the PC and displayed on the screen of the PC monitor.

3. Precise analogue signal conversion using AVR analogue-to-digital converter with 10-bit resolution.

4. All data acquired by the system is logged into a database for future reference with date and time of sam-

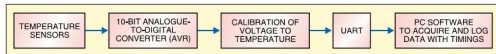


Fig. 1: Block diagram of eight-channel data acquisition and logging system



Fig. 2: Author's prototype

and other physical parameters on a centralised machine whereby one can monitor and control the remote devices. The AVR microcontroller-based system described here does the same job of acquiring the analogue data and sending it to a remote terminal for monitoring.

Fig. 1 shows the block diagram of the eight-channel data acquisition and logging system using AVR microcontroller and Fig. 2 shows the author's prototype. The key features of this system are:

1. The software is user-friendly and written in VB 6.0.
2. Data is acquired through serial

pling.

5. The internal analogue-to-digital conversion (ADC) channels of the AVR are used to acquire real-time data in the form of analogue signal. The data is sent to the PC via UART channel.

Circuit description

Fig. 3 shows the circuit of the eight-channel

data acquisition and logging system using AVR. At the heart of the circuit is ATmega32 AVR microcontroller from Atmel.

The ATmega32 microcontroller has 32 kB of flash program memory, 2 kB of SRAM, internal analogue-to-digital converter (ADC) with 10-bit resolution, internal EEPROM and full-duplex UART channel. This data logger uses ADC channels of the AVR to acquire real-time data in the form of analogue signal and sends this data to the PC via UART channel.

Vcc (pin 10) and AVcc (pin 30) of the AVR are connected to +5V for operation. By default, this AVR works



with the internal RC oscillator at 1 MHz. Here, fuse bits of the AVR are set to operate an external oscillator. We have used an external stable crystal oscillator to run at a frequency of 16 MHz.

The AVR has internal power-on reset facility. Resistor R2 (10-kilo-ohm), capacitor C5 (10µF) and switch S1 make up the external reset circuitry. Switch S1 allows you to reset the system at run time.

Analogue reference voltage pin V_{REF} (pin 32) is connected to the variable terminal of the 10-kilo-ohm preset. Using this preset, you can adjust the ADC reference voltage.

We have used all the eight channels of the 10-bit ADC for acquiring the analogue voltage proportional to the environmental temperature of temperature sensors.

The in-built UART channel of ATmega32 is used to send the current data to the host PC. UART works on 9600 bauds per second. The length of RS-232 serial cable is tested for operation up to 10 metres but it should work up to 15 metres.

Data acquisition and logging

Temperature sensor. Temperature sensor LM335 from National Semiconductors has been used in this project. Its pin details are shown in Fig. 4.

LM335 has a breakdown voltage directly proportional to absolute temperature at 10 mV/°K with less than 1-ohm dynamic impedance. The device operates over a current range of 400 µA to 5 mA with virtually no change in performance. LM335 can be used in